THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

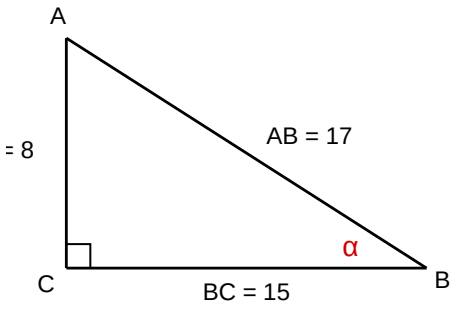
ANSWER KEY & MARKING SCHEME — MATHEMATICS · CLASS 10 · SET 05

Max. Marks: 80 · The Mastermind Coaching Classes, Jaipur · Alternate correct methods to be given full credit

SECTION – A (Multiple Choice Questions · 20 × 1 = 20) **20 questions · 20 marks**

Q1–Q18 are multiple-choice questions. Q19–Q20: a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option.

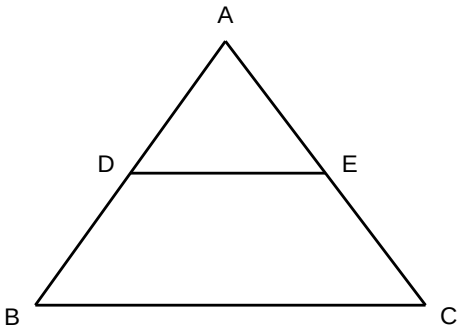
1. From the given figure, the value of cos α is



(a) $\frac{8}{17}$ (b) $\frac{8}{15}$ (c) $\frac{15}{17}$ (d) $\frac{15}{8}$

Ans: (c) cos α = BC/AB = 15/17.

2. In the figure DE ∥ BC, D and E are the midpoints of AB and AC. If BC = 14 cm, then DE is



(a) 14 cm (b) 7 cm (c) 28 cm (d) 3.5 cm

Ans: (b) By the midpoint theorem DE = ½ BC = 7 cm.

3. The HCF and LCM of 24, 36 and 60 are respectively

(a) 6, 360 (b) 12, 720 (c) 24, 180 (d) 12, 360

Ans: (d) $24 = 2^3 \cdot 3$, $36 = 2^2 \cdot 3^2$, $60 = 2^2 \cdot 3 \cdot 5$ ⇒ HCF = $2^2 \cdot 3 = 12$, LCM = $2^3 \cdot 3^2 \cdot 5 = 360$.

4. The probability that a randomly chosen day of a non-leap year is in the month of April is

(a) $\frac{30}{365}$ (b) $\frac{1}{12}$ (c) $\frac{31}{365}$ (d) $\frac{30}{366}$

Ans: (a) April has 30 days ⇒ 30/365.

5. Which of the following points CANNOT lie on the *y*-axis?

(a) (0, −3) (b) (−3, 0) (c) (0, 0) (d) (0, 7)

Ans: (b) A point on the y-axis has x = 0; (−3, 0) has x ≠ 0.

6. The value of $(\sqrt{5} + \sqrt{2})^2 - 2\sqrt{10}$ is

(a) $7 + 4\sqrt{10}$ (b) 3 (c) $5 + 2\sqrt{10}$ (d) 7

Ans: (d) $5 + 2 + 2\sqrt{10} - 2\sqrt{10} = 7$.

7. The number $5 \times 7 \times 11 + 7 \times 5$ is

(a) prime (b) irrational

(c) composite (d) neither prime nor composite

Ans: (c) $= 35(11 + 1) = 35 \times 12$, composite.

8. The fares of a taxi are ₹ 30 for the first km and ₹ 12 for each additional km. The fare for 9 km is

(a) ₹ 126 (b) ₹ 120 (c) ₹ 138 (d) ₹ 108

Ans: (a) $30 + 8 \times 12 = 126$.

9. The pair $5x - 2y = 4$ and $10x - 4y = k$ is coincident for

(a) $k = 8$ (b) $k = 4$ (c) $k = 2$ (d) $k \neq 8$

Ans: (a) $5/10 = 2/4 = 4/k \Rightarrow k = 8$.

10. The value of $(1 + \tan^2\theta)\cos^2\theta$ is

(a) $\sin^2\theta$ (b) 1 (c) $\tan^2\theta$ (d) 0

Ans: (b) $\sec^2\theta \cdot \cos^2\theta = 1$.

11. The distance of the point $(-6, 8)$ from the origin is

(a) 14 (b) 2 (c) 10 (d) 6

Ans: (c) $\sqrt{(36 + 64)} = 10$.

12. A bag has 3 red, 5 white and 2 blue balls. One ball is drawn at random. The probability that it is not white is

(a) $\frac{1}{5}$ (b) $\frac{3}{10}$ (c) $\frac{2}{5}$ (d) $\frac{1}{2}$

Ans: (d) Not white $= 5/10 = 1/2$.

13. If $2x + y = 8$ and $x - y = 1$, then $x + y$ equals

(a) 3 (b) 4 (c) 7 (d) 5

Ans: (d) $3x = 9 \Rightarrow x = 3$, $y = 2 \Rightarrow x + y = 5$.

14. If the HCF of 45 and 75 is 15, then their LCM is

(a) 225 (b) 150 (c) 450 (d) 1125

Ans: (a) $45 \times 75/15 = 225$.

15. Two dice are thrown together. The probability that the sum is 7 is

(a) $\frac{1}{12}$ (b) $\frac{5}{36}$ (c) $\frac{1}{6}$ (d) $\frac{7}{36}$

Ans: (c) 6 outcomes $\Rightarrow 6/36 = 1/6$.

16. If $\tan\theta = \frac{5}{12}$, then $\sin\theta$ equals

(a) $\frac{12}{13}$ (b) $\frac{5}{13}$ (c) $\frac{5}{12}$ (d) $\frac{13}{5}$

Ans: (b) Hypotenuse 13 $\Rightarrow \sin\theta = 5/13$.

17. The equations $x + 2y = 4$ and $2x + 4y = 12$ have

(a) one solution (b) two solutions

(c) infinitely many solutions (d) no solution

Ans: (d) $1/2 = 2/4 \neq 4/12 \Rightarrow$ parallel.

18. The common difference of the AP in which $a_{18} - a_{14} = 32$ is

(a) 4 (b) 8 (c) −8 (d) −4

> **Ans: (b)** 4d = 32 ⇒ d = 8.

19. **Assertion (A):** Assertion (A): The mid-point of the segment joining (2, 5) and (6, 9) is (4, 7).
Reason (R): Reason (R): The mid-point of (x_1, y_1) and (x_2, y_2) is $((x_1 + x_2)/2, (y_1 + y_2)/2)$.

> **Ans: (a)** Both true; R is the formula used.

20. **Assertion (A):** Assertion (A): The probability of an impossible event is 0.
Reason (R): Reason (R): The probability of a sure event is 1.

> **Ans: (b)** Both are true, but R does not explain A.

SECTION – B (Very Short Answer Questions · 5 × 2 = 10) — 5 questions · 10 marks

21. Find the HCF and LCM of 396 and 1080 by the prime factorisation method.

> $396 = 2^2 \times 3^2 \times 11$ and $1080 = 2^3 \times 3^3 \times 5$ (1); HCF = $2^2 \times 3^2 = 36$; LCM = $2^3 \times 3^3 \times 5 \times 11 = 11880$ (1).

O R

Three bells toll at intervals of 9, 12 and 15 minutes. If they toll together at 8 a.m., when will they next toll together?

> LCM(9, 12, 15) = 180 min (1) ⇒ 11:00 a.m. (1).

22. A pole 6 m high casts a shadow 4 m long. At the same time a tower casts a shadow 50 m long. Find the height of the tower.

> 6/4 = h/50 (1) ⇒ h = 75 m (1).

23. In ΔABC, DE ∥ BC with AD = 4 cm, DB = 6 cm and AE = 6 cm. Find EC.

> AD/DB = AE/EC ⇒ 4/6 = 6/EC (1) ⇒ EC = 9 cm (1).

O R

Two similar triangles have areas 49 cm² and 81 cm². If a side of the smaller is 14 cm, find the corresponding side of the larger.

> Ratio = 7/9 (1) ⇒ 14 × 9/7 = 18 cm (1).

24. A rope tied to the top of a vertical pole makes 30° with the level ground and is 24 m long. Find the height of the pole.

> sin 30° = h/24 (1) ⇒ h = 12 m (1).

25. Find the value of y if the distance between (2, y) and (10, −3) is 10 units.

> $64 + (y + 3)^2 = 100$ (1) ⇒ $y + 3 = \pm 6$ ⇒ y = 3 or −9 (1).

SECTION – C (Short Answer Questions · 6 × 3 = 18) — 6 questions · 18 marks

26. Prove that $5 - 3\sqrt{2}$ is irrational, given that $\sqrt{2}$ is irrational.

> Assume $5 - 3\sqrt{2} = p/q$ (1) ⇒ $\sqrt{2} = (5q - p)/(3q)$ rational (1), contradiction ⇒ $5 - 3\sqrt{2}$ is irrational (1).

27. Solve: $3x - 5y = -1$ and $x - y = -1$.

> x = y − 1 (1); 3y − 3 − 5y = −1 ⇒ y = −1 (1), x = −2 (1).

28. Find the sum of all three-digit natural numbers which are multiples of 11.

> 110, 121, …, 990: n = (990 − 110)/11 + 1 = 81 (1½); S = 81/2 × (110 + 990) = 44550 (1½).

OR

The 9th term of an AP is 0. Prove that its 29th term is twice its 19th term.

> $a + 8d = 0 \Rightarrow a = -8d$ (1); $a_{29} = 20d$ (1), $a_{19} = 10d \Rightarrow a_{29} = 2a_{19}$ (1).

29. Find the ratio in which the *y*-axis divides the line segment joining A(−4, −6) and B(10, 12). Also find the point of division.

> $x = 0 \Rightarrow (10k - 4)/(k + 1) = 0 \Rightarrow k = 2/5$, ratio 2 : 5 (1½); $y = (2 \times 12 + 5 \times (-6))/7 = -6/7 \Rightarrow (0, -6/7)$ (1½).

OR

The vertices of a triangle are (0, 0), (6, 0) and (3, 3√3). Show that it is equilateral.

> Sides: 6, $\sqrt{(9 + 27)} = 6$, 6 (3) ⇒ equilateral.

30. If $\sin\theta = \frac{7}{25}$, find the value of $\sec\theta + \tan\theta$.

> $\cos\theta = 24/25$ (1); $\sec\theta = 25/24$, $\tan\theta = 7/24$ (1); sum = 32/24 = 4/3 (1).

31. A bag contains 4 red, 5 black and 6 green balls. A ball is drawn at random. Find the probability that it is (a) red (b) not green (c) red or black.

> Total 15. (a) 4/15 (1). (b) 9/15 = 3/5 (1). (c) 9/15 = 3/5 (1).

SECTION – D (Long Answer Questions · 4 × 5 = 20) — 4 questions · 20 marks

32. Prove that the ratio of the areas of two similar triangles is equal to the square of the ratio of their corresponding sides.

> Given ΔABC ~ ΔPQR; to prove $ar(ABC)/ar(PQR) = (AB/PQ)^2 = (BC/QR)^2 = (CA/RP)^2$ (1). Draw AM ⊥ BC, PN ⊥ QR (½). $ar(ABC)/ar(PQR) = (½ \cdot BC \cdot AM)/(½ \cdot QR \cdot PN) = (BC \cdot AM)/(QR \cdot PN)$ (1). ΔAMB ~ ΔPNQ (AA) ⇒ AM/PN = AB/PQ (1). Also AB/PQ = BC/QR ⇒ ratio = (BC/QR)·(AB/PQ) = $(AB/PQ)^2$ (1). Similarly equal to the other ratios (½).

33. A fraction becomes 9/11 if 2 is added to both numerator and denominator, and 5/6 if 3 is added to both. Find the fraction.

> $(x + 2)/(y + 2) = 9/11 \Rightarrow 11x - 9y = -4$ (1½); $(x + 3)/(y + 3) = 5/6 \Rightarrow 6x - 5y = -3$ (1½). Solve: x = 7, y = 9 (1½). Fraction 7/9 (½).

OR

Draw the graphs of $2x + y = 6$ and $2x - y = 2$ and find the area of the triangle formed with the *y*-axis.

> Tables (2); graph (1); solution (2, 2) (1); y-intercepts 6 and −2 ⇒ base 8, height 2 ⇒ area 8 sq. units (1).

34. From the top of a 75 m high lighthouse the angles of depression of two ships on opposite sides are 30° and 45°. Find the distance between the ships. (Take √3 = 1.732)

> Distances: $75/\tan 30° = 75\sqrt{3}$ (2); $75/\tan 45° = 75$ (2). Total $= 75(\sqrt{3} + 1) = 75 \times 2.732 = 204.9$ m (1).

35. Prove that $\frac{\sin\theta - 2\sin^3\theta}{2\cos^3\theta - \cos\theta} = \tan\theta$.

> LHS $= \sin\theta(1 - 2\sin^2\theta)/[\cos\theta(2\cos^2\theta - 1)]$ (2); $1 - 2\sin^2\theta = 2\cos^2\theta - 1$ (2) ⇒ $\tan\theta$ (1).

OR

If $\sin(A - B) = ½$ and $\cos(A + B) = ½$, $0° < A + B \le 90°$, $A > B$, find A and B.

> $A - B = 30°$ (1½), $A + B = 60°$ (1½) ⇒ $A = 45°$, $B = 15°$ (2).

SECTION – E (Case Study Based Questions · 3 × 4 = 12) — 3 questions · 12 marks

36. A cricket academy holds a prize draw. 40 tickets numbered 1 to 40 are placed in a drum and one is drawn at random.

(i) Find the probability that the number is a multiple of 5.

> 8/40 = 1/5 (1).

(ii) Find the probability that the number is a perfect square.

> 1, 4, 9, 16, 25, 36 ⇒ 6/40 = 3/20 (1).

(iii) Find the probability that the number is prime and odd.

> Primes ≤ 40: 12; odd ones: 11 ⇒ 11/40 (2).

OR

(iii) A ticket numbered 10 is drawn and kept aside. Find the probability that the next ticket drawn is even.

> 39 left; evens left: 19 ⇒ 19/39 (2).

37. A farmer plots a field on a grid in which one unit is 10 m. The corners are A(1, 2), B(7, 2), C(7, 10) and D(1, 10).

(i) Find the length AB in metres.

> AB = 6 units = 60 m (1).

(ii) Find the length of the diagonal AC in metres.

> AC = √(36 + 64) = 10 units = 100 m (1).

(iii) Find the coordinates of the point where the diagonals meet.

> Midpoint of AC = (4, 6) (2).

OR

(iii) A well is at the point P on AC with AP : PC = 3 : 2. Find P.

> P = ((3×7 + 2×1)/5, (3×10 + 2×2)/5) = (23/5, 34/5) (2).

38. A cinema hall has 20 rows of seats. The first row has 15 seats and each next row has 2 more seats than the row before it.

(i) Write the first term and common difference of the AP.

> a = 15, d = 2 (1).

(ii) How many seats are in the 10th row?

> 15 + 9 × 2 = 33 (1).

(iii) Find the total seats in the hall.

> S_{20} = 10(30 + 38) = 680 (2).

OR

(iii) Which row has 45 seats?

> 15 + 2(n − 1) = 45 ⇒ n = 16 (2).